

Lab #3: DC-DC boost prep, diodes and transistors

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Table partners:

#1. Experiment description and Prelab

##1.1 Description

In this lab, I will study the behaviour of transistors and diodes. Diodes are essentially a one way valve for current, while transistors are basically electrically controlled switches (i.e controlled by voltage drop, or current). We will be using a MOSFET transistor in this lab, which operates as a voltage controlled switch.

In my analysis, I have to test the I-V relation of a diode according to the setup in the prelab. This will be compared to the ideal case, where forward biased current is infinite or extremely large. Then, I will build a circuit using this diode to produce a truncated sine wave, where the bottom half of the wave is a constant 0, while the top half still follows regular wave behaviour.

Then, I will analyze a MOSFET in a circuit. This involves building a MOSFET circuit to control an LED light, where the LED will light up when the MOSFET is forward biased. Specifications for this circuit will be in its corresponding section later in this writeup.

I will also be familiarizing myself with standard safety and lab procedures for handling MOSFETs, including the use of grounding/shock bracelets.

##1.2 Prelab

prelab task 1: designing a circuit & measurement procedure to measure the I-V relation of a diode for the prelab figure (see below).

```
In [3]: InlineImage('diode.png', width=12, height=12, caption= "Figure 1, prelab ideal diode I-V relation", fs=12);
```

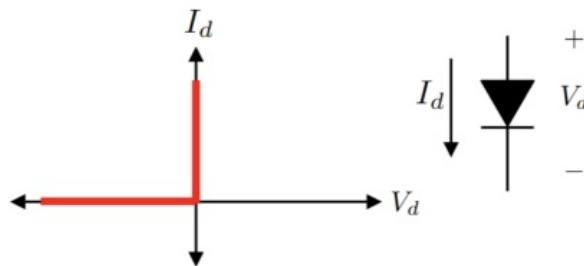


Figure 1: The current-voltage relation of an ideal diode: when forward biased ($V_d > 0$) the current is infinite, other the current is zero. Inset: circuit symbol

Figure 1, prelab ideal diode I-V relation

The I-V relation is essentially just the graph showing V on the x axis and I on the Y axis. Our diode will not be ideal, meaning conductivity is not ideal; current will not increase instantly all the way to infinity. As discussed in the tutorial, there has to be some 'threshold' potential or electromotive force accross the diode that must be met for current to start increasing. We will also expect that the diode cannot perfectly prevent backflow of current, so I would expect to see some negative current when voltage is negative. I also remember in class that the magnitude of voltage gets too big in the backwards direction, the diode may fail to block the backwards flowing current, causing a steep negative increase in current.

Below is my circuit design to measure this relation:

```
In [7]: InlineImage('diodeess.png', width=12, height=12, caption= "Figure 2, circuit diagram to measure the I-V relation");
```

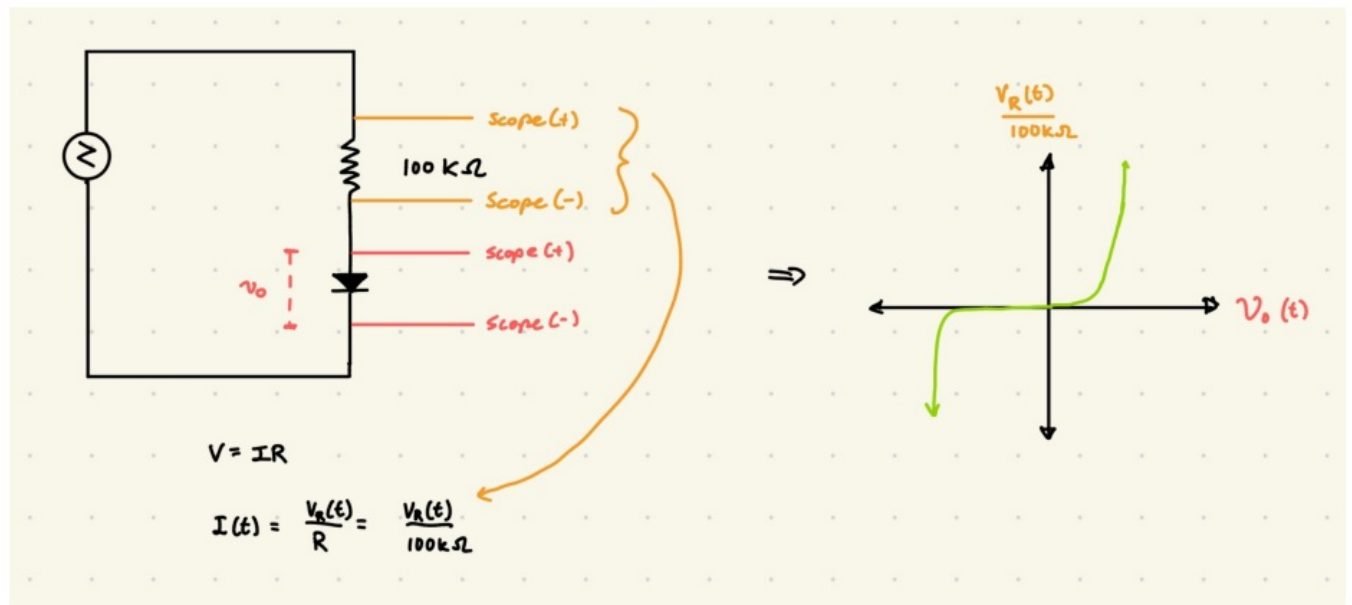


Figure 2, circuit diagram to measure the I-V relationship of a diode using a $100\text{ k}\Omega$ resistor, with expected I-V relation

Brief explanation: my circuit essentially uses a triangle wave to vary the voltage linearly, (I will decide the amplitude later during experimentation), and look at the voltage across the diode. In order to find current, I will look at the voltage across the resistor, and with the resistance value, I can find the current in the circuit at any time. Theoretically, I can then match the current at some time t_0 with voltage across the diode at that same t_0 and graph this relationship.

Prelab task 2:

Designing a switch circuit for an LED using a MOSFET with a 5V power supply. Current should be within 10% of 20mA, and voltage drop should be 2.5V.

Since my circuit only consists of a resistor and led in series, with a MOSFET as a switch, and there are only voltage drops in the loop over the resistor and LED, we can find the value of the resistor needed. Since the power supply is 5V, and the voltage drop across the LED should be 2.5V, we should also have a 2.5V voltage drop across the resistor. For $V = IR$, we have $R = \frac{V}{I} = \frac{2.5\text{V}}{20\text{mA}} = 125\Omega$. So the resistor should have a value of 125Ω .

Below is my diagram:

In [5]: `InlineImage('mossfett.png', width=12, height=12, caption= "Figure 3, circuit diagram of my MOSFET LED setup. Ser`

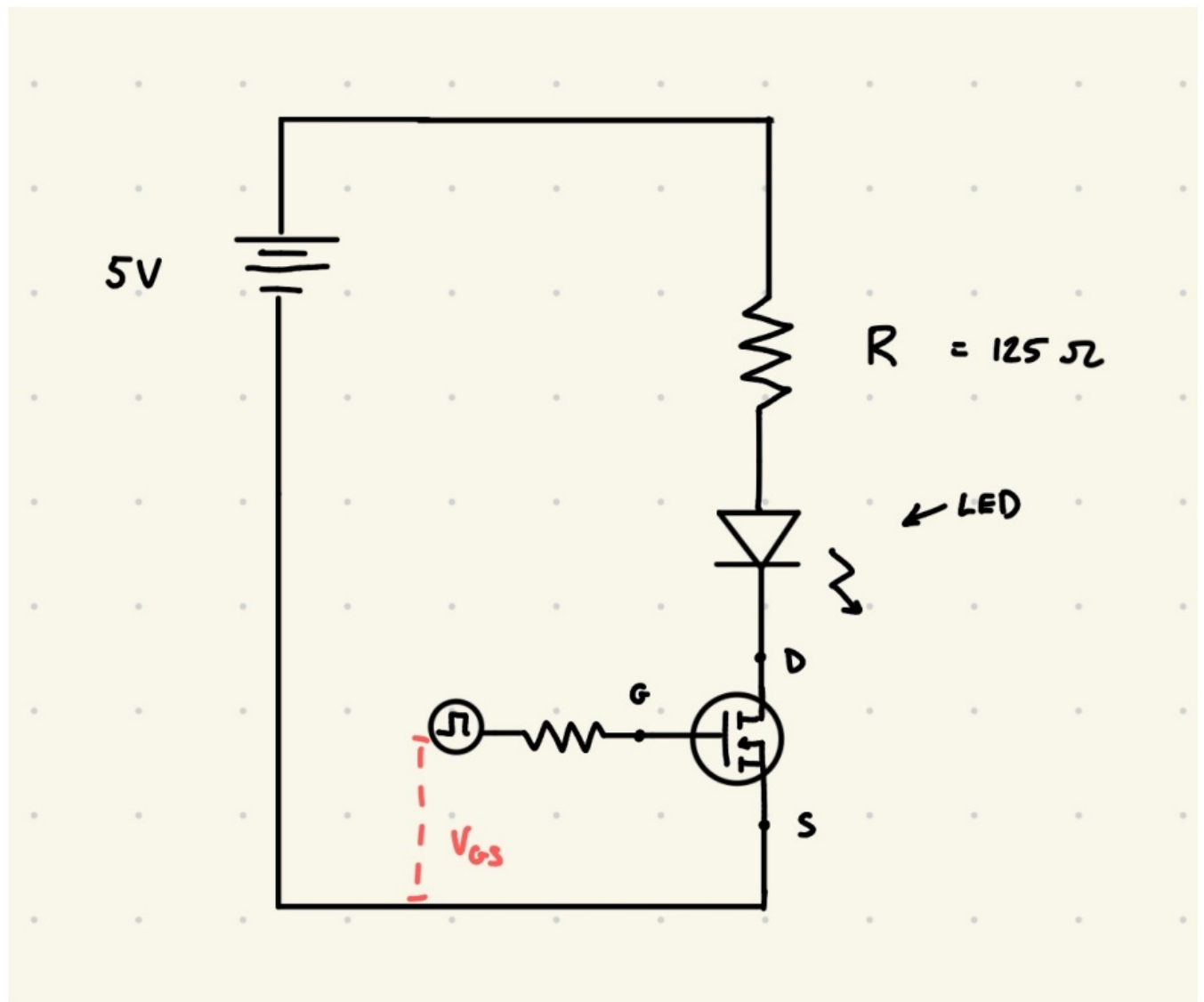


Figure 3, circuit diagram of my MOSFET LED setup. See above for resistor value reasoning.

Note: I assumed we would use a square wave to simulate the switch being turned on or off; once V_{GS} reaches a certain value, the mosfet will allow or disallow current from D to S. Actually now that I think about it, a triangle wave could work too especially for finding the value for which current will flow through from D to S.

#2. Experiment

##2.1 Ideal vs non ideal diodes

First, I will build the circuit to obtain the I-V relation for the diode. I built the circuit from figure 2:

In [11]: `InlineImage('dioders.png', width=16, height=10, caption= "Figure 4, circuit with resistor and diode in series p`

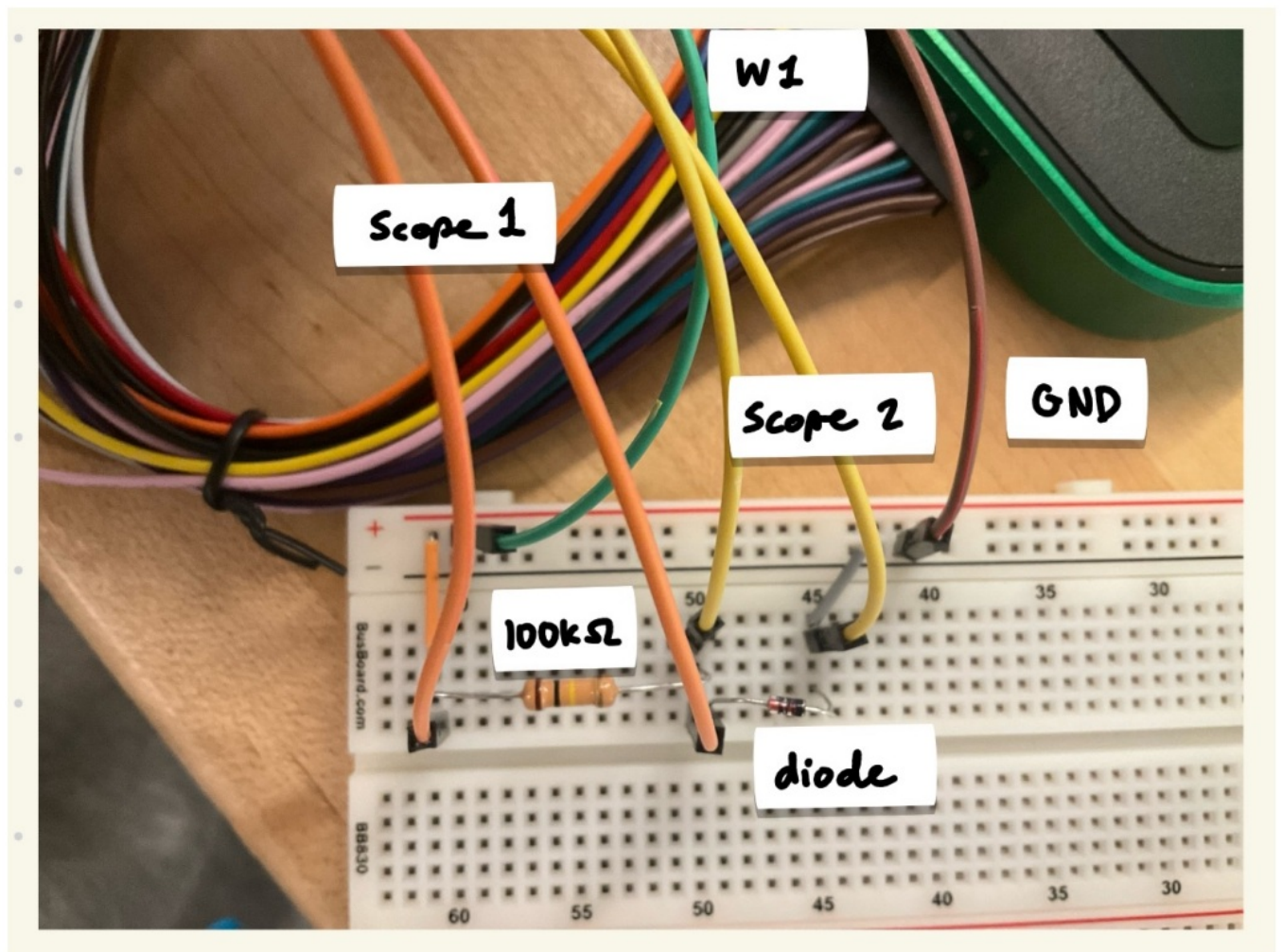


Figure 4, circuit with resistor and diode in series powered by a triangle wave. See figure 2. for corresponding circuit diagram.

The circuit works, and I get a waveform. **TS** However, I don't think I am seeing the full voltage change. Hence, I will change the frequency of the input triangle wave. Below is my initial waveform that I see, prior to any changes.

In [12]: `InlineImage('initialwave.png', width=16, height=10, caption= "Figure 5, Initial waveform with diode and resistor")`

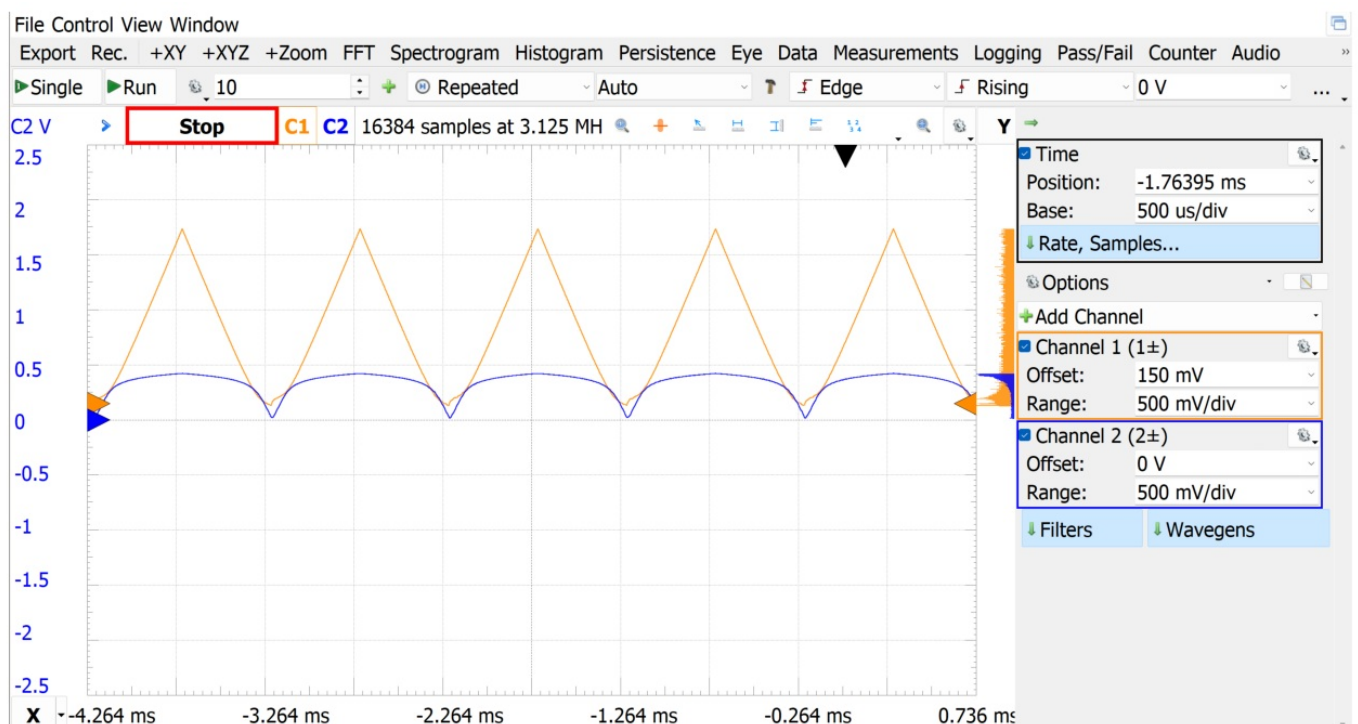


Figure 5, Initial waveform with diode and resistor circuit, using default settings for a triangle wave of 1V amplitude and 1V offset.

In the meantime, I will set up a graph. I have divided the voltage across the resistor by 100 000 Ω , and plotted it against the voltage across the diode. Here is what the initial graph looks like:

In [14]: `InlineImage('wierdone.png', width=16, height=10, caption= "Figure 6, Initial graph of current as a function of voltage across the diode.")`

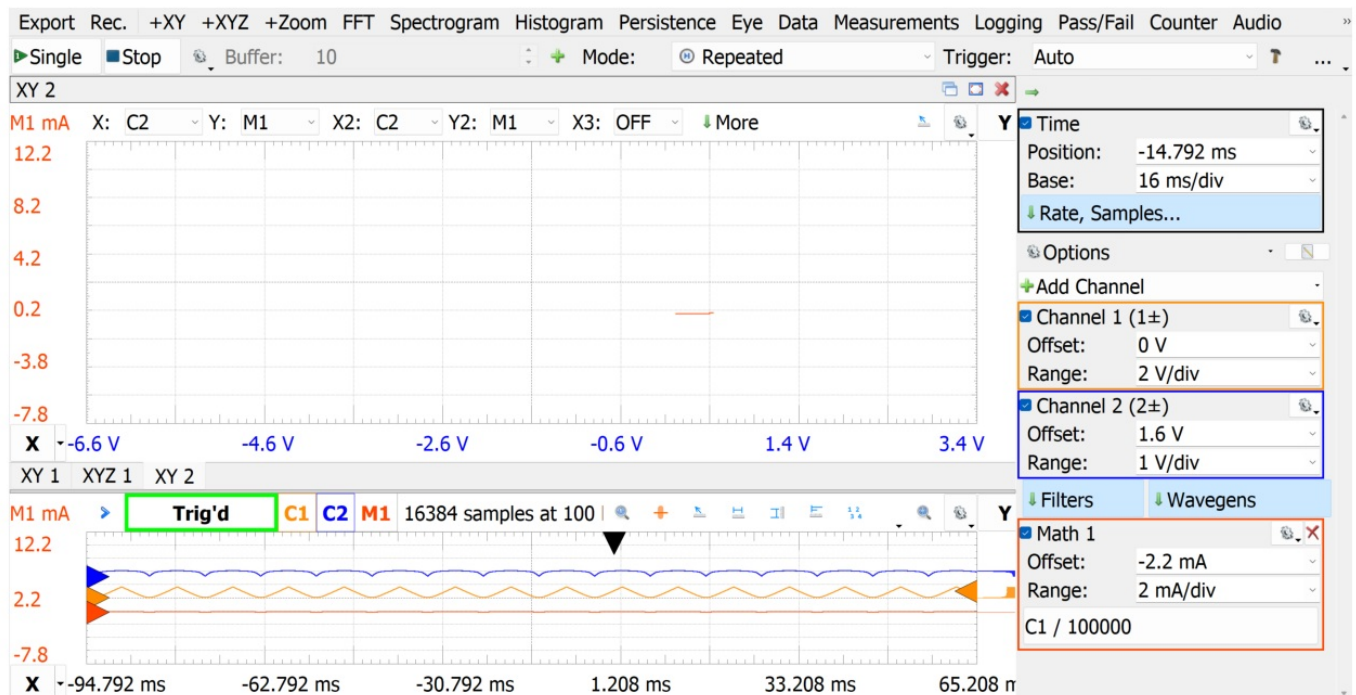


Figure 6, Initial graph of current as a function of voltage across the diode.

My resistor value is $98.3 \pm \frac{0.1}{\sqrt{3}} k\Omega$, as measured by my voltmeter.

TS ok, so my graph looks like a horizontal line... which is not what it is supposed to look like. I will try the following:

1. change the window settings for y axis scale
2. double check the math

Results:

1. Did not really change anything. Still a straight line.
2. I tried dividing it by 100 instead of 100 000, and now I see the positive portion of the graph I sketched in figure 2. This is the same as keeping the value in $k\Omega$, but then the final value has units in mA rather than A. Actually, this makes sense now; the waveform software axis is in mA by default, and I was giving it an input of μA by dividing voltage by 100 000. Also, I now see a tiny lift at the end of the graph, which is probably because my values were just 1000x too small to see a noticeable change!!.

After troubleshooting, I corrected the division to the value I measured for the resistor rather than 100.

In [19]: `InlineImage('goodgraph.png', width=16, height=10, caption= "Figure 7, graph of current vs voltage across a diode.")`
`InlineImage('goodset.png', width=6, height=10, caption= "Figure 8, waveform settings for triangle wave for \ngon")`

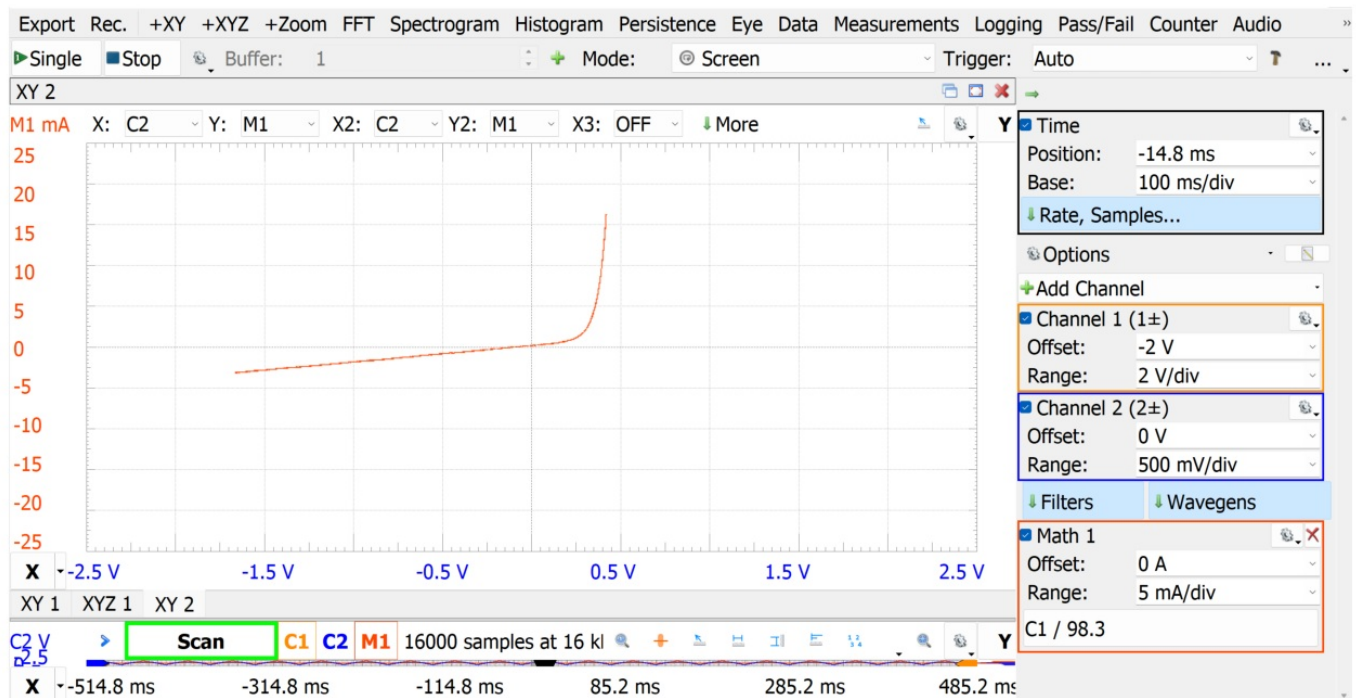


Figure 7, graph of current vs voltage across a diode. Centered at 0,0.

Channel 1 (W1)	
☒ Enable	Simple Idle:
Type:	$\wedge$ Triangle
Frequency:	20 Hz
Period:	50 ms
Amplitude:	2 V
Offset:	0 V
Symmetry:	50 %
Phase:	0 °

Figure 8, waveform settings for triangle wave for good visual of the waveform.

Now I will plot this in python here directly.

```
In [34]: #x = np.linspace(0,2*np.pi,50) # Creates an array of 50 equally spaced floats between 0 and 2pi
#sinewave = np.sin(x) # Computes the sine of each element in 'x' and stores the result in 'sinewave'

fig,ax = plt.subplots(figsize=(8,8)) # Generates a figure of size 8x8 along with an axis object

data = pd.read_csv("IVrelations.csv")
V = data['Channel 2 (V)'] #Columns in pandas dataframes can be indexed by their header
I = data['Math 1 (A)']
ax.grid(True)

ax.plot(V, I, color='r', linewidth=2, zorder=1, label='Line plot') # Line plot 'sinewave' vs. 'x' in red with 2
#ax.scatter(t, V, color='k',s=2, zorder=2, label='Scatter plot') # Scatter plot 'sinewave' vs. 'x' in black with
ax.set_xlim(-0.8,0.8)
# #ax.set_xticks([0,np.pi/2,np.pi,3*np.pi/2,2*np.pi]) # Matplotlib will automatically generate ticks, but you c
# ax.set_xticklabels(['0','$\pi/2$','$\pi$','$3\pi/2$','$2\pi$']) # Rather than display fractions of pi as deci

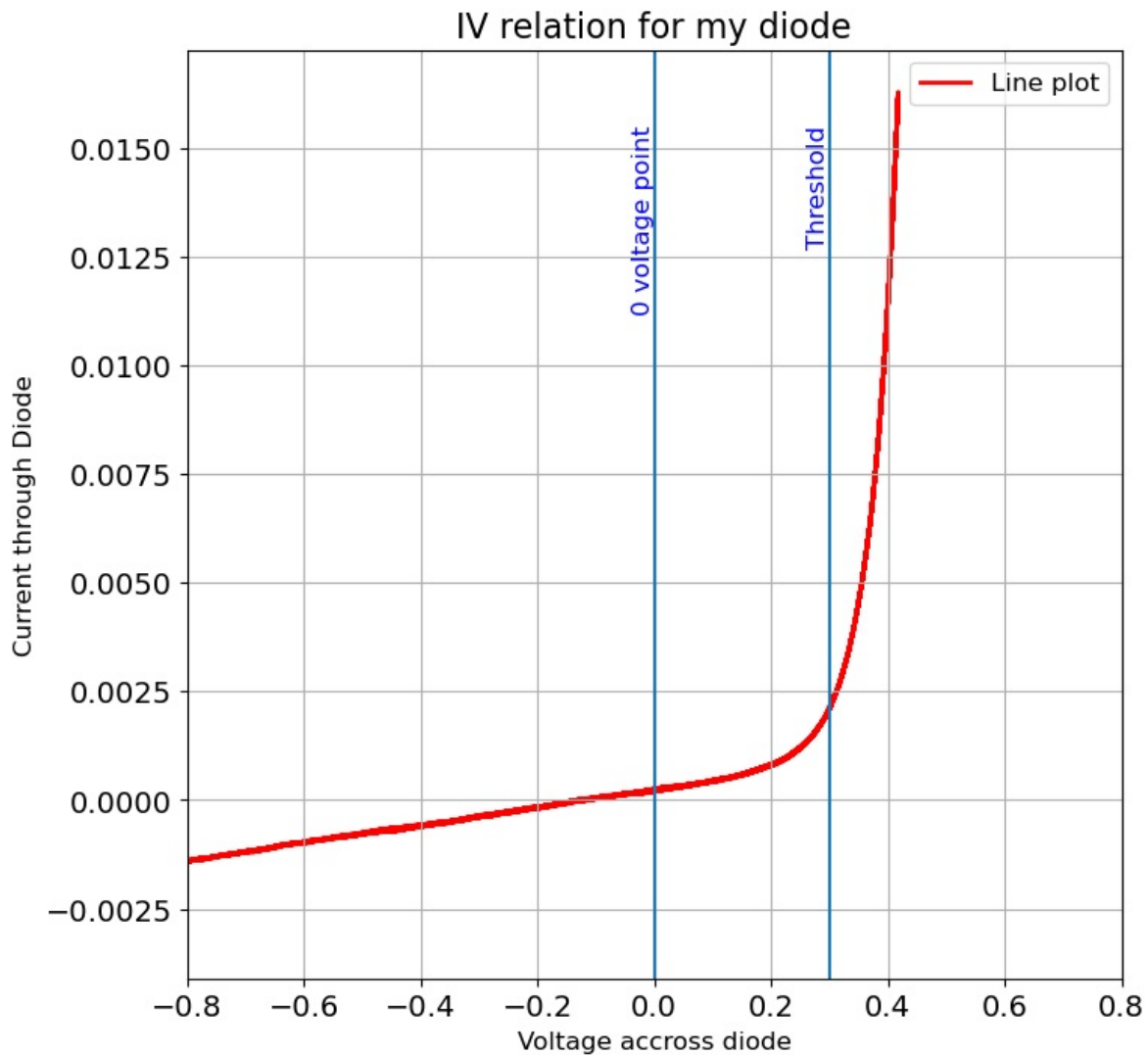
# Make sure your fonts are large enough to be legible!
ax.tick_params(axis='x', labelsz=14)
ax.tick_params(axis='y', labelsz=14)

# Make sure to label your axes!
ax.set_ylabel('Current through Diode',fontsize=12)
ax.set_xlabel('Voltage accross diode',fontsize=12)

ax.set_title('IV relation for my diode', fontsize=16)

ax.text(0.3, ax.get_ylim()[1]*0.9, 'Threshold', color='b', rotation=90, va='top', ha='right', fontsize=12)
ax.text(0, ax.get_ylim()[1]*0.9, '0 voltage point', color='b', rotation=90, va='top', ha='right', fontsize=12)
ax.axvline(0.3)
ax.axvline(0)
ax.legend(loc='upper right', fontsize=12) # We can include a simple legend like this - the labels assigned to e

Out[34]: <matplotlib.legend.Legend at 0x1d603fded0>
```



This is my plot for the I V relation. I have labelled the rough threshold point at which the current starts to spike, as well as the portion at 0 voltage.

Conceptual question 1

This plot has a few differences from the ideal case as I showed earlier in this writeup:

1. In the negative voltage range, the voltage is not perfectly 0. This means that the diode is not perfectly stopping the current backflow, which makes sense. In the ideal case, the diode should have infinite resistance for backwards current but of course that is unlikely in reality.
2. At the 0 voltage point, there is already a bit of current passing through the diode. This also means that the resistance is not perfect; the diode 'leaks' a little bit.
3. In the positive voltage range, there is a gradual increase rather than a sudden spike to infinite current. This is due to the fact that in reality, the propagation of anything cannot be instant. As such, the current must gradually increase (although this is still on a super fast time scale for us). The diode is also not a perfect conductor with ideal properties of 0 resistance; so current must increase gradually.

4. There seems to be a "threshold" point at which the current starts to increase at an exponential rate. This is around 0.3Volts for my diode; after this point, current increases at a super fast rate.

The voltage drop across a forward biased diode can even be slightly negative as seen on my graph; when there is a very small positive current, the voltage across the diode is slightly negative and very small. Maybe this is real physics at work, where the diode is slightly leaky...? This may also be due to the error in the scope. However, I calculated this; the fractional uncertainty for the resistor applied to this has magnitude of 10^{-7} . That is not enough to make this due to error, unless the scope itself has a super large error too.

##2.2 diode based rectifier

for this part, we are trying to recreate a sinusoidal waveform with essentially the negative voltage portion cut off. Basically, we are just trying to see proof that the diode blocks backwards current, which will occur when the voltage drop is negative (since current flows from higher to lower potential).

The setup will be very similar to the setup for the previous section. However, we will mostly just change the location of the scope probes:

```
In [35]: InlineImage('circuit1.png', width=12, height=10, caption= "Figure 9, circuit diagram of diode and resistor in series to measure the single direction sinusoid.")
InlineImage('circuit11.png', width=12, height=10, caption= "Figure 10, circuit setup of diode and resistor in series to measure the single direction sinusoid.")
```

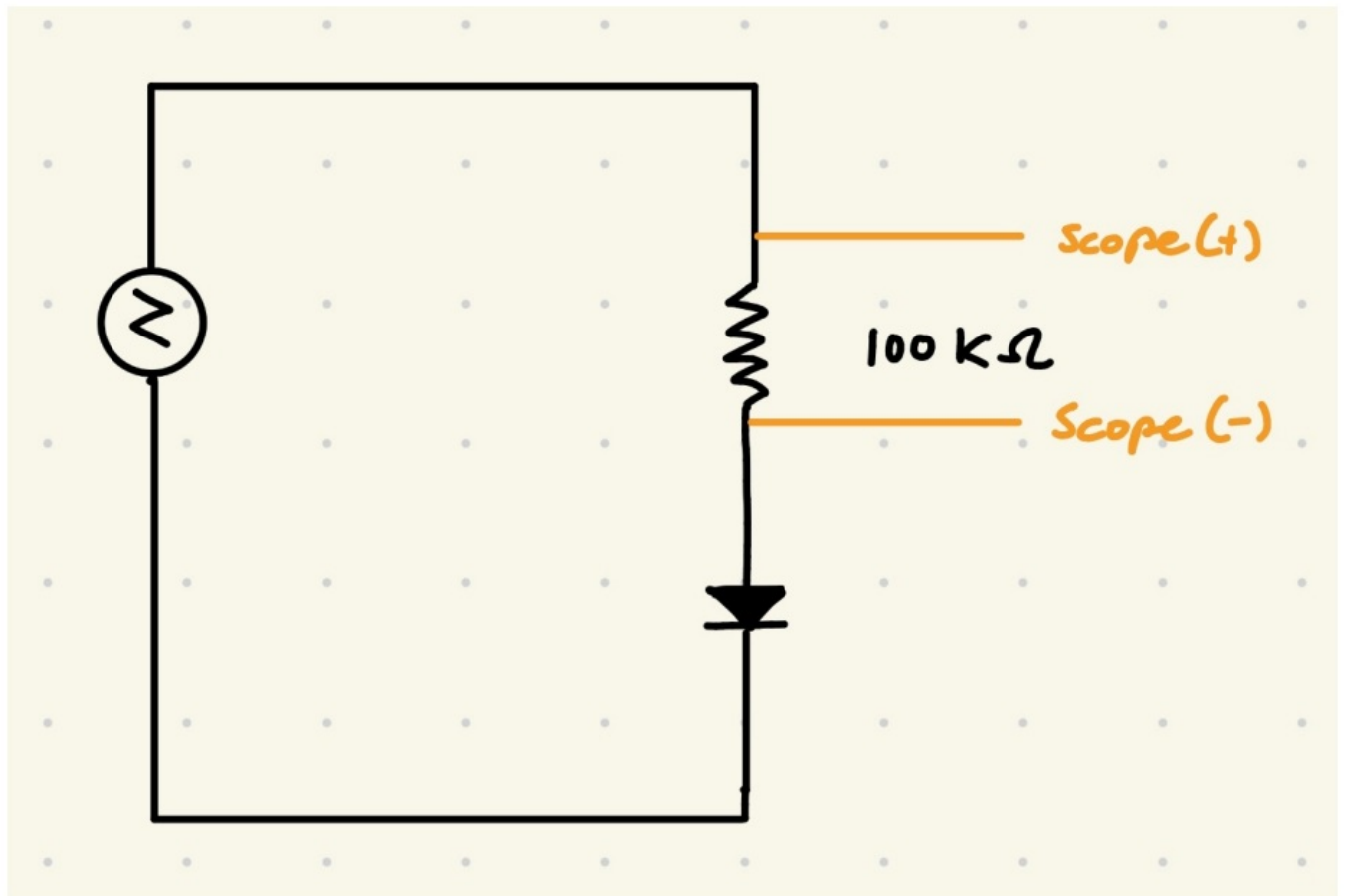


Figure 9, circuit diagram of diode and resistor in series to measure the single direction sinusoid.

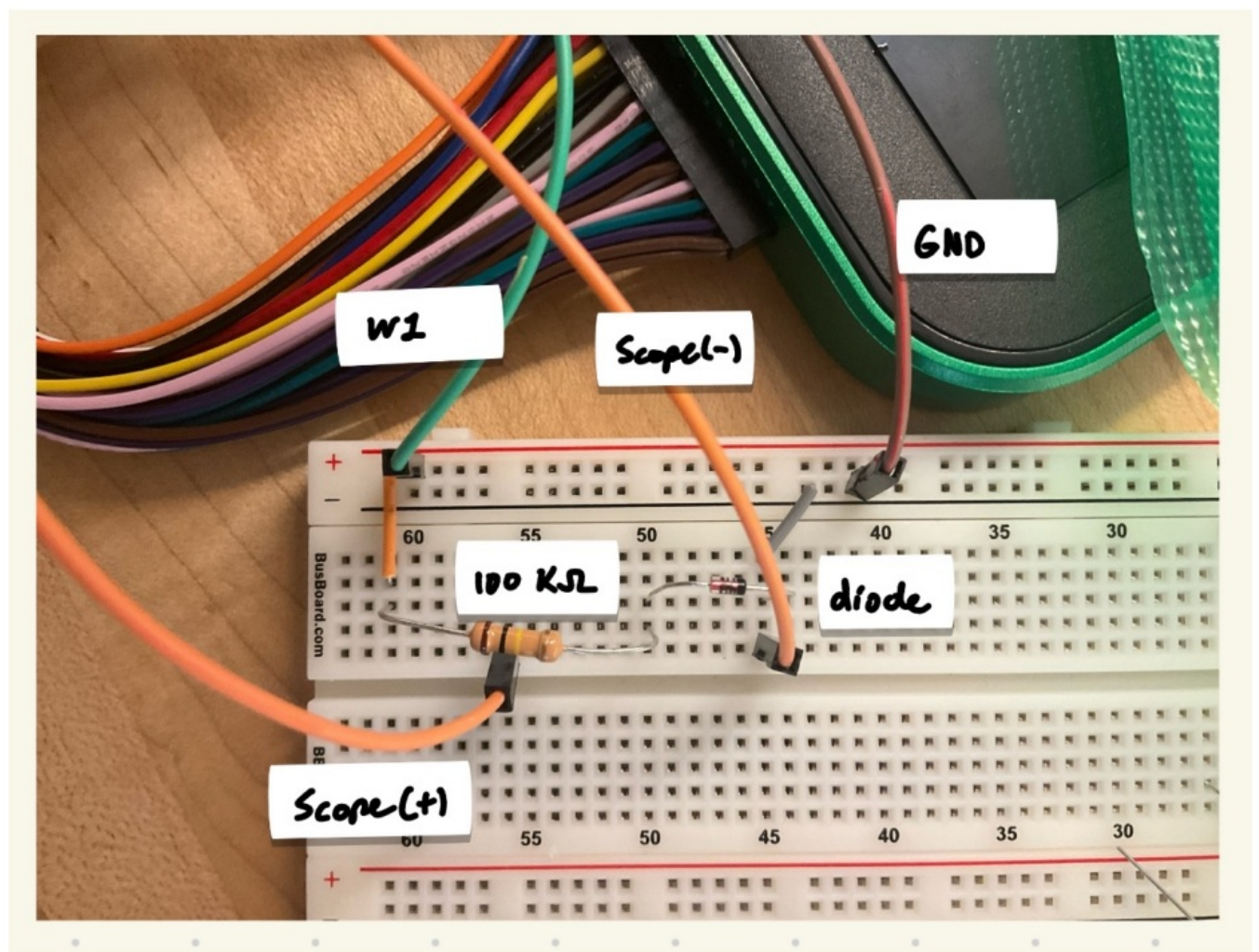


Figure 10, circuit setup of diode and resistor in series to measure the single direction sinusoid.

My waveform looks a bit funky... but I am not sure why. The part where it should be flat actually has a negative value...? see graph below:

In [36]: `InlineImage('funkywave.png', width=12, height=10, caption= "Figure 11, voltage over the resistor with negative "`



Figure 11, voltage over the resistor with negative voltage/backflow current

TS I have a few things in mind. There is obviously current flowing backwards, maybe the diode is not properly blocking the current? Here are a few things I tried:

1. Testing the voltage across the diode only - this gave me a relatively normal looking waveform - a sine wave with a positive amplitude, and a lower negative amplitude. This is expected as we know the diode is not 100% effective in real life, and so I can conclude the diode seems to work fine.
2. Test 1 gave me an idea. I have the diode behind the resistor right now, maybe the curved bottom area for negative voltage was propagating backwards over to the resistor?? If I switch the diode to go ahead of the resistor, maybe it will work?

I also noticed here that the diode has a right way and a wrong way; the black part on the diode should be at lower potential than the red portion.

Ok, so I have tried switching the parts around- now, I get a regular waveform! See below:

In [38]: `InlineImage('halfsine.png', width=12, height=10, caption= "Figure 12, Successful half-sinusoid for voltage over`
`InlineImage('sinesine.png', width=8, height=10, caption= "Figure 13, settings for the 1 V sinusoidal voltage source`

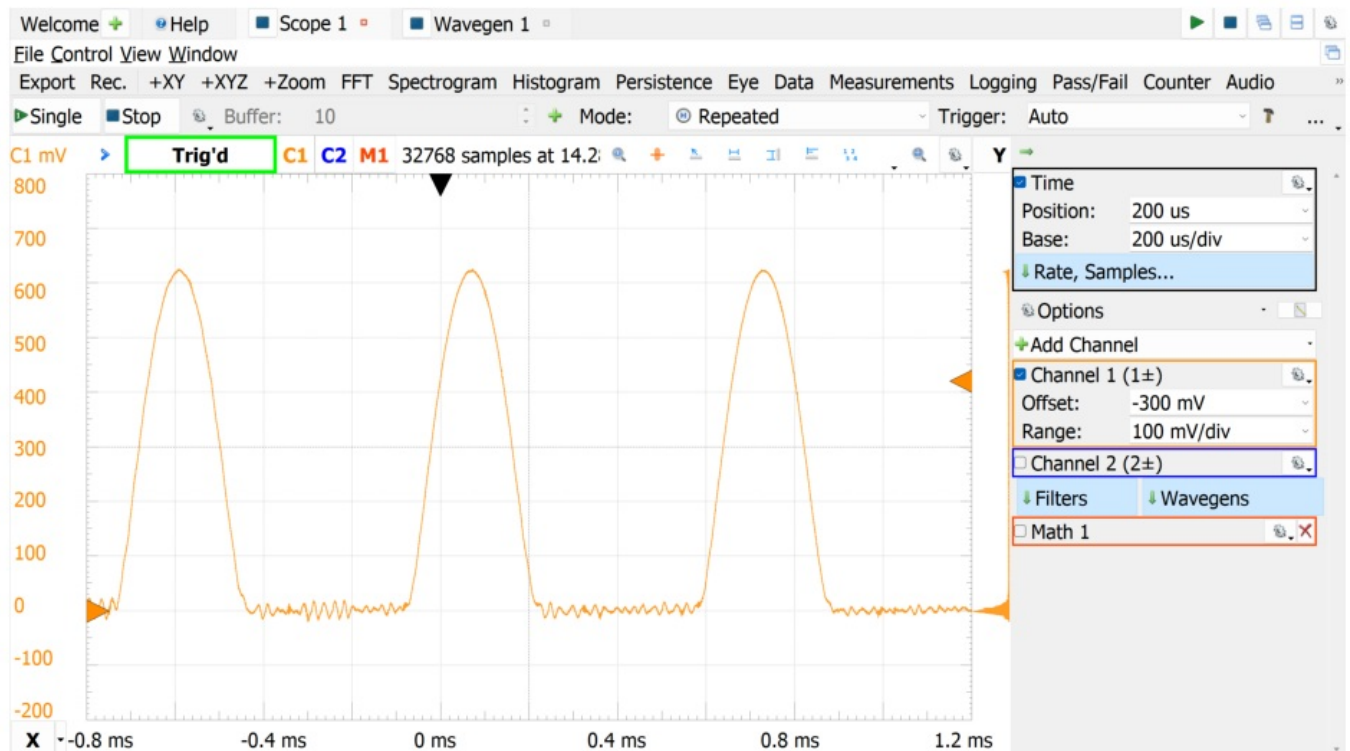


Figure 12, Successful half-sinusoid for voltage over the resistor in series with a diode. Sinusoidal 1V waveform source.

Type:	$\sim$ Sine
Frequency:	1.515151515 kHz
Period:	660 μ s
Amplitude:	1 V
Offset:	0 V
Symmetry:	50 %
Phase:	0 $^{\circ}$

Figure 13, settings for the 1 V sinusoidal voltage source.

Note: Period was obtained by approximating off of the example waveform in the prelab document.

##2.3 MOSFET LED circuit

In building the circuit, I realized we don't have 25 ohm resistors. As such, I will use 4 100ohm in parallel, and place that in series with another 100 ohm. This is not exactly necessary, as theoretically this only serves to control/fine tune the amount of current through the LED. As long as the resistance is in a particular range, I don't think there is a risk of blowing the LED; it will only make it brighter or dimmer. However, for the sake of my calculations (and since I have access to the resistors), I will just make my resistance in the circuit 125 Ohms. :)

Another change I have made in this circuit compared to the first circuit is the removal of the resistor that separated the wave source and the MOSFET. This was because prior to the lab, I was sort of confused about the MOSFET and how it differed from a NPN transistor. I realize now that for a MOSFET, since there is no current dependence whatsoever, I don't need a resistor to tune the current coming from the wave source to the transistor. A MOSFET depends on the voltage only, and as such, a resistor for the wave source is pointless (pretty much).

See below for my updated circuit diagram and circuit!

In [8]: `InlineImage('finaldiagram.png', width=12, height=10, caption= "Figure 14, Circuit diagram for a MOSFET controll`

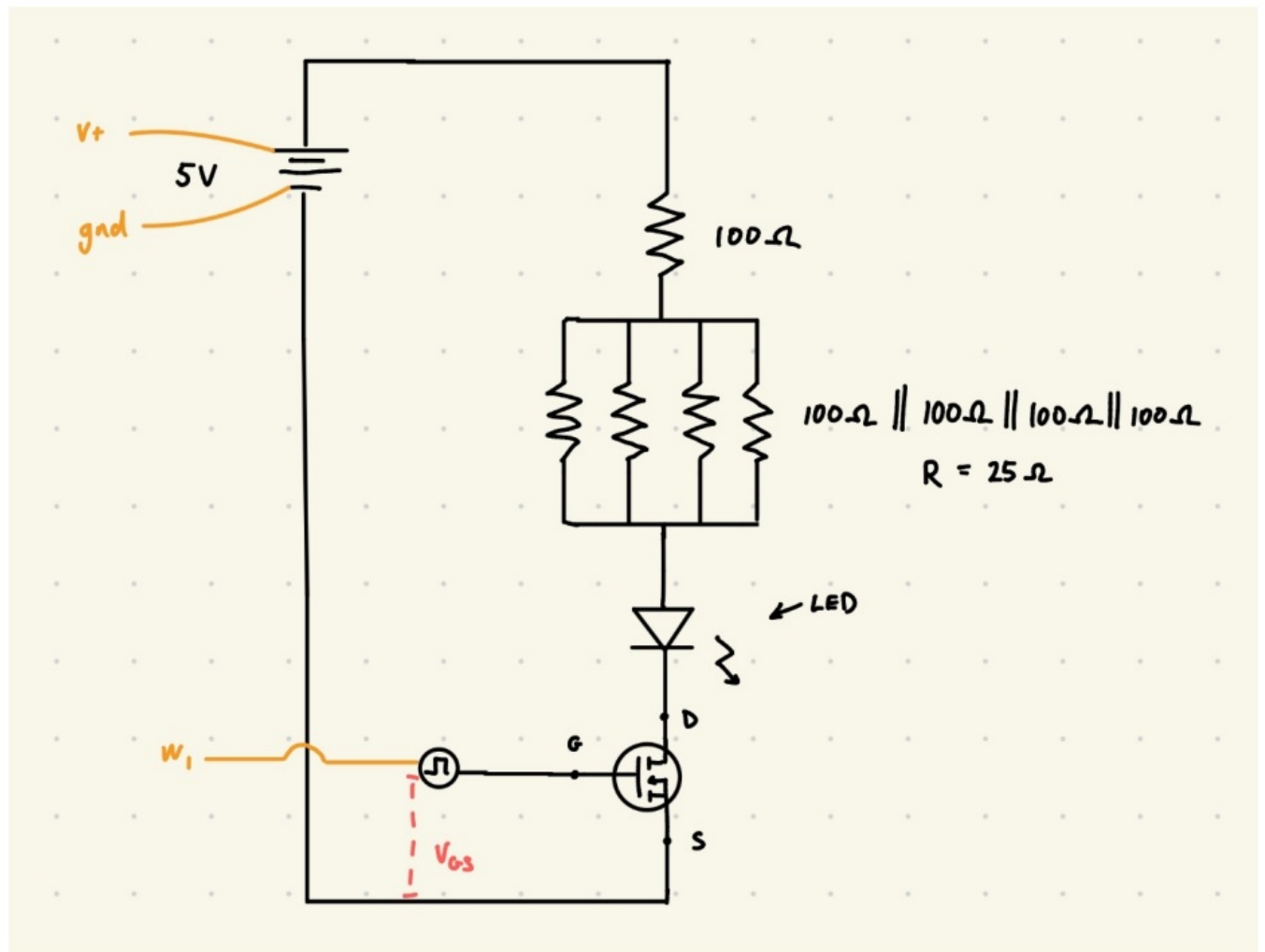


Figure 14, Circuit diagram for a MOSFET controlled LED using a current-controlling resistance of 125 Ω . Resistance obtained by use of 4 parallel 100 Ω in series with another 100 Ω

I recall from the lecture that the LED is basically just a diode that emits light. That means that it has a polarity to let current through. I have taken extra care to ensure that the longer leg of the LED, which is the positive one, is oriented at higher potential so the LED can light up properly. This will be essential as the power source is DC meaning the current only ever flows in one direction. This could make it difficult to troubleshoot later on.

In [10]: `InlineImage('ledworking.png', width=12, height=10, caption= "Figure 15, circuit of 125  $\Omega$  in series with i`

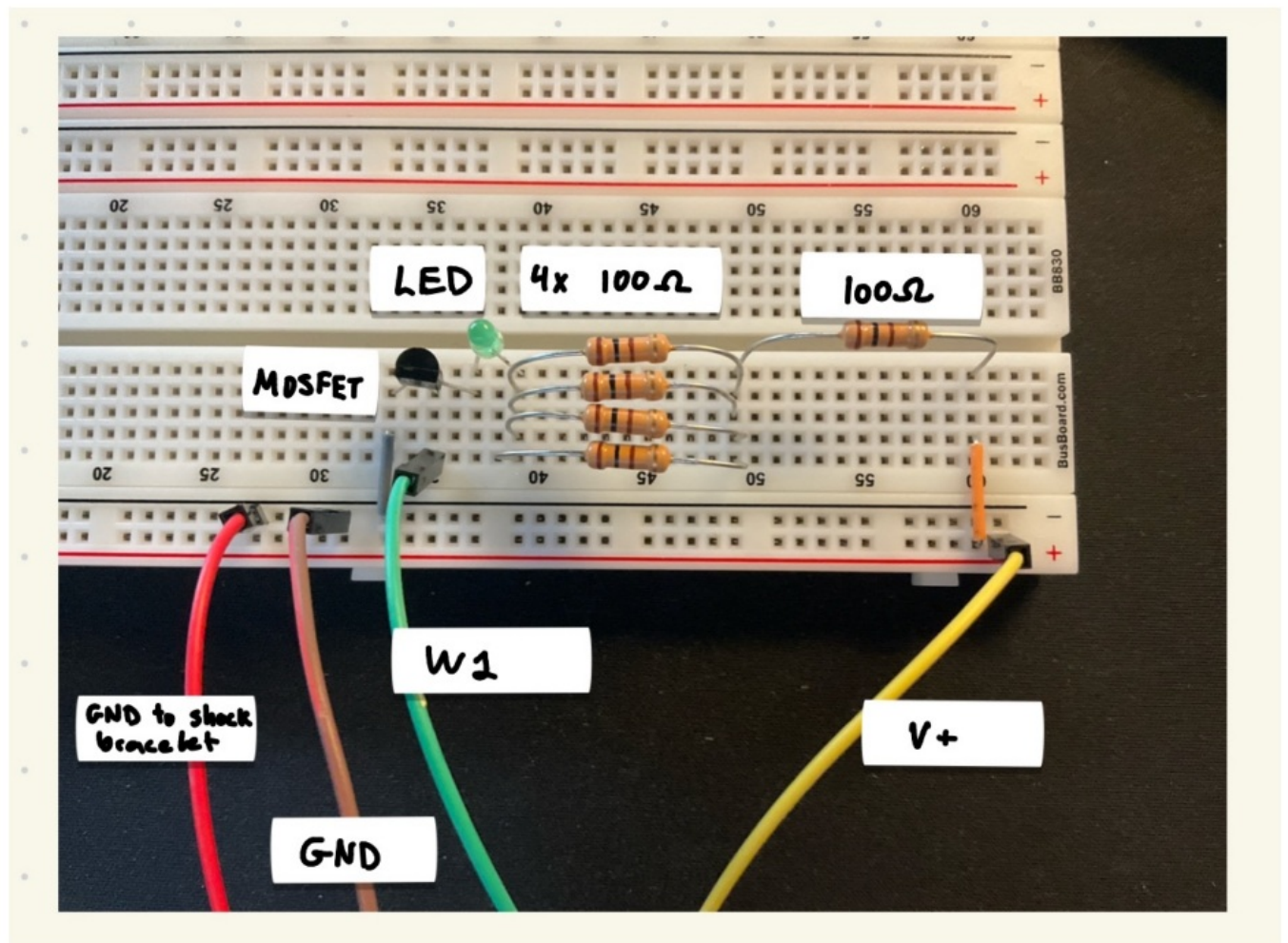


Figure 15, circuit of $125\ \Omega$ in series with an LED and MOSFET transistor. LED is powered by 5V DC source, and MOSFET gate is controlled by 1V square wave.

It works! I can adjust the magnitude of the source square wave to adjust the brightness of the LED. I can also adjust the frequency/period of the source square wave to make it blink faster or slower as I like. I also tested the LED with a sinusoidal source wave, and gave a very interesting gradual blinking pattern. With this, I also found that the threshold potential for the MOSFET to have a forward bias/on mode was approx 3V. I found this value by changing the offset of the source waveform. Above 3V, the LED is constantly on. Below, the LED will be off.

Conceptual question 2

For confirming if the MOSFET is operational:

There should only be a forward voltage drop over the diode between the drain and source pins because as seen in the equivalent circuit, the transistor essentially functions off of a voltage dependent current source. This is the MOSFET's 'gate' which only allows current to flow through to one side. Essentially, an ideal current source has infinite resistance so that the current it produces is the only current passing through it; as such, when we make it voltage controlled, it becomes a one way flow valve, where how open it is will be controlled by the voltage drop between the gate and ground. We need this one-way behaviour for both voltage drop (since current should flow from positive to negative potential) and current for the MOSFET to be considered operational; this is exactly the theory behind the MOSFET. Testing this is also how I found out how to orient the MOSFET in the circuit (ie. which terminal is positive) first try and get the circuit to work with no problems.

The combinations of Gate, Drain and Source pins should measure to have high resistances (i measured in the mega ohms) because ideally they should not have any current flow between the pins. In the equivalent circuit for the MOSFET, we can see that there is no actual connection between the three terminals apart from Drain to Source. However, due to the current source, we should also find a super high level of resistance. The MOSFET wants to control how much current is let through based ONLY on the voltage drop between gate and ground. Any other current would be unideal. Because of this high resistance, we don't have any current running through and lighting the LED when we don't want, which is the whole point of using a MOSFET to regulate it.

Hence, these conditions must be true for a MOSFET to be considered effective.

Below is an image of the equivalent circuit to a MOSFET.

In [11]: `InlineImage('equivalent.png', width=12, height=10, caption= "Figure 16, equivalent circuit design for a MOSFET,`

MosFet equivalent circuit

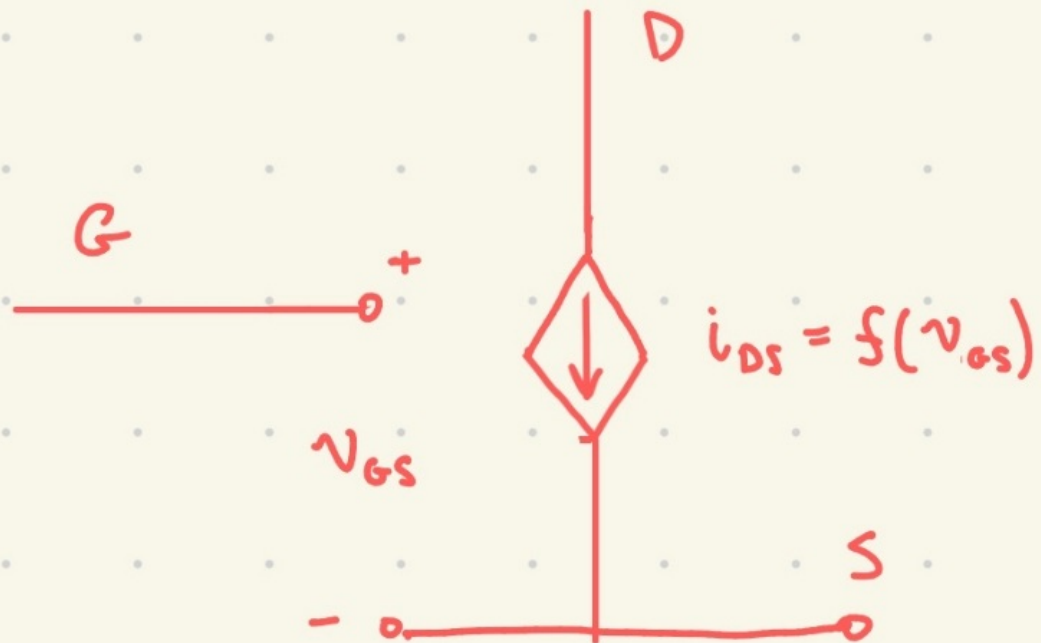


Figure 16, equivalent circuit design for a MOSFET, involving a voltage dependent current source.

Conceptual question 3

Removing the connections to gate and replacing it with an open wire resulted in the LED turning off. This is to be expected, since there is no longer a significant voltage drop between gate and ground. If the wire were unconnected and the LED were somehow on, then we could expect the air to be at higher potential than ground...? wait but then the circuit would not function since air touches the ground too haha. I think that now that the wire is no longer connected to anything, there could be energy stored with nowhere to go, similar to a capacitor. I think in the tutorials there was something said about this and how we can damage the MOSFETs due to this stored charge/energy. (and so we should take extra care near static and i guess any sort of residual charges)

Grabbing the open end of the wire and touching the circuit resulted in a few things:

touching the circuit prior to the Gate pin of the MOSFET would make the LED light up. This is likely because at the gate pin, the actual 'gate' of the MOSFET exists. Hence, with the 5V DC source giving voltage to the circuit prior, there is higher potential than ground! So the open wire measures this and relays it to gate, allowing current to flow and power the LED.

touching the circuit on gate and afterwards resulted in no light up. This is because the MOSFET has not had the gate open, so there is no potential since no current from the 5V DC source, so the gate cannot open; this is like charging a dead battery with a dead battery.

touching the wire to me did nothing. This is because I was grounded with the wrist strap. However, removing the wrist strap caused the LED to light up. I think this is because it is detecting any charge that I have on me, and is measuring a potential which activates the MOSFET. OK, so now I know that the MOSFET is extremely sensitive to any external charges, and even simply touching it to a non-grounded conductor can effect circuit function. As such, in order for the circuit to behave as I desire, I need to make sure the mosfet is always correctly connected. This is also necessary to prevent damage to the MOSFET, in case a large amount of static builds up and has nowhere to go.

From the above behaviour, I can estimate that the input resistance is probably very high for this circuit/MOSFET. This is because for any small amount of current, we can cause a change in the circuit behaviour as long as the voltage drop is enough. Given $V = IR$; I should be negligibly small in comparison to V and R since MOSFET is a voltage dependent switch.

For a given V , if we make I super small (since that's all we need), we have a large R value. This also makes sense with the high resistance values measured by the oscilloscope in checking MOSFET functionality.

Essentially, we don't need a very high current; in fact we barely need any. So long as the voltage 'threshold' is met, we can turn the MOSFET transistor on. This highlights the MOSFET's function as a voltage controlled device rather than a current controlled device like other transistor types. As such, MOSFETs can reduce the power dissipation that comes with current flowing through wires. (may be more

practical for modern devices).

#3. Conclusion

In this lab, I primarily looked at the function of MOSFETs and a little bit about Diodes.

First, I observed the real life V-I relationship for a diode, where I saw that the current does not jump suddenly to infinity in the forwards biased case as we would expect in theory. We found that this is due to the fact that current must be continuous in reality and cannot mimic an instant increase to some value. This is also because there is a threshold value for the voltage across a diode before we can observe a large current increase. This might be because the charges do not have enough energy to travel across the potential until some threshold value.

I then looked at a diode based rectifier, where I found it was important to place the diode before the resistor to prevent any current backflow. Otherwise, the resistor will still have a slight negative portion of voltage in the sinusoidal waveform. The diode also has a correct orientation that we need to be careful of; troubleshooting diode function can be as simple as flipping it around in the circuit.

Then, I looked at MOSFET circuits and built one to control an LED. I found that MOSFETs also have a potential value for voltage that must be met in order to turn on/turn the LED on. For my particular one, this threshold was around 3V. I also discovered that MOSFETs are not 'all or nothing' like some responses (for example our neurons). We can tune them with a voltage level to control the current through, since the current is proportional to the voltage from gate to ground. As such, we can control devices that behave differently with varying currents, such as this LED - where brightness changes. I also found that MOSFETs are very sensitive to external charge or static, and that to prevent charge buildup (that may be fatal to the MOSFET), I should always keep it connected to something in the circuit. To prevent unwanted circuit behaviour, I should do the same. I will also remember to take care ensuring I am grounded to not damage the MOSFET accidentally.

General notes:

- Devices such as these are polarized, and I should make sure to test polarity before using them in my circuits.
- Always connect MOSFETS
- LEDs are also diodes, they just emit light on top of standard diode function.
- It is NOT ALWAYS the voltage difference between gate and ground that causes the MOSFET to turn on/off - I only referred to that throughout this lab as that was the case for my circuit setup. The Source pin of the MOSFET was connected to ground. However, I realize that this may not always be the case in future circuits.

```
In [2]: # comments in Python start with a "#" and will not be executed - well commented code is easier for graders to u
# Importing modules that will be necessary for the tutorial
import numpy as np # numpy is a general math/array manipulation library that is almost always used in any scienc
import matplotlib.image as mimg # matplotlib.image is used for displaying images in python
import matplotlib.pyplot as plt # matplotlib.pyplot is used for general plotting and is very similar to matlab
import pandas as pd # pandas is a data manipulation library, we will use it to import data from excel files
print('All necessary modules imported!')
```

All necessary modules imported!

```
In [4]: def InlineImage(img, width=8, height=8, caption=None, fs=24):
# Author: BG
# Renders an image with a caption underneath using matplotlib
# Requires matplotlib.pyplot to be imported as plt and matplotlib.image to be imported as mimg
# img - string, the location of an image file
# width, height - numbers, the dimensions of the figure the image is displayed in. Original aspect ratio of
# caption - string, the caption to be displayed beneath the image
# fs - number, the font size of the caption
fig,ax = plt.subplots(figsize=(width,height))
ax.imshow(mimg.imread(img))
ax.set_xlabel(caption,fontsize=fs)
ax.tick_params(
    which='both',
    bottom=False,
    top=False,
    left=False,
    labelleft=False,
    labelbottom=False)
ax.spines["top"].set_visible(False)
ax.spines["left"].set_visible(False)
ax.spines["right"].set_visible(False)
ax.spines["bottom"].set_visible(False)
```